

Can LLMs Write Actually Good Jokes?

From alignment-tax metrics to culturally grounded humor evaluation

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Why “good jokes” are hard for LLMs

Easy version

- ▶ Complete familiar joke formats.
- ▶ Produce joke-shaped text.
- ▶ Increase randomness or edginess.

Hard version

- ▶ Be specific to the situation.
- ▶ Understand audience and culture.
- ▶ Find a point of view.
- ▶ Select the right kind of surprise.

Motivation

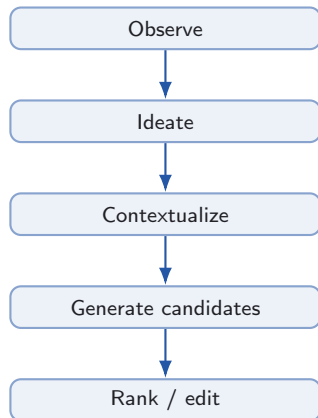
A punchline can be surprising because it is clever, but also because it is random, incoherent, or offensive. So we care about whether LLM humor is not just surprising, but **actually funny**.

Humor as a structured creative process

Inspiration: HumorSkills

Rather than asking a model to “just be funny,” HumorSkills decomposes humor generation into steps:

- ▶ observe concrete details
- ▶ generate humorous angles
- ▶ infer social or narrative context
- ▶ generate many candidates
- ▶ rank and filter outputs



How this connects to our project

Our current midterm anchor measures distributional signals like surprisal, entropy, and semantic distance. The broader goal is to connect those signals to **humor quality**.

Formulating humor as an LLM problem

Generation setting

$$y_i \sim P_{\theta_m}(y \mid x, c)$$

- ▶ x : setup, prompt, image, or news/context
- ▶ y_i : generated joke or punchline candidate
- ▶ m : model condition, e.g. base / instruct / safety-finetuned
- ▶ c : decoding, style, audience, or genre condition

Core question

- ▶ How does alignment shift the joke-output distribution?
- ▶ Do those shifts correspond to better or worse humor?

Two evaluation layers

- ▶ Distributional proxies: surprisal, entropy, semantic distance
- ▶ Quality score: $f_\phi(x, y)$ for funniness, coherence, specificity

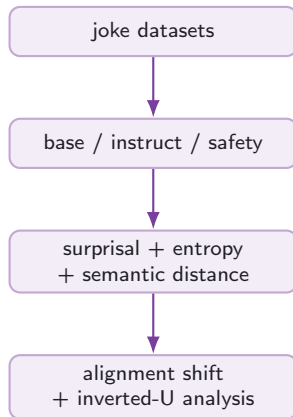
Midterm anchor: alignment-tax pipeline

What we have now

- ▶ Joke datasets: `r_jokes`, `one_liner`
- ▶ Model comparison: base, instruct, safety-finetuned
- ▶ Metrics: surprisal, entropy, semantic distance
- ▶ Analysis: distributional shifts and inverted-U tests

Midterm role

- ▶ Controlled, measurable preliminary layer
- ▶ Gives us a quantitative anchor before humor-quality evaluation



Preliminary result: distributional shift

Current finding

The initial pipeline lets us test whether alignment changes joke-related distributional metrics. Preliminary results suggest that alignment changes the distribution, but not necessarily through simple compression.

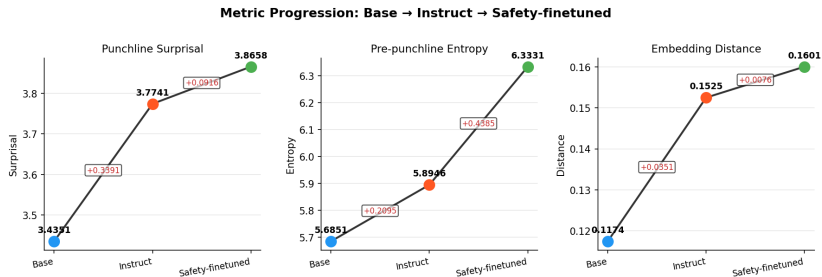


Figure: preliminary analysis of surprise shifts.

Next steps: connecting metrics to humor quality

What we are adding

A humor-quality track: can structured generation produce jokes that are not merely surprising, but specific, culturally grounded, and funny?

Immediate next steps before final

- ▶ Add LLM/human judging to connect metrics with perceived humor.
- ▶ Filter or separate clean vs. offensive jokes to reduce confounds.
- ▶ Test stronger LLM/VLM systems for qualitative humor generation.
- ▶ Explore HumorSkills-style or agentic generation as a future extension.

Surprise is measurable. Humor is situated.